

nect these as per polarity of the 12V battery. Overcharge status of 14.4V is set with the help of preset VR3. Overcharge status is indicated by LED2.

We have to switch off S1 to protect the battery from overcharging. During normal charging, no LED (LED1 or LED2) will glow. If S1 is off, the rectifier circuit formed by diodes D3 and D4 will be disabled, which, in turn, will stop further charging of the battery. Lower limit of the battery is set at 11.3V with the help of preset low level of the battery, which will be indicated by LED1. Switch S1 has to be closed to restart the charging of the battery. Load should be disconnected when battery voltage is lower than 11.3V and mains voltage is not there.

Oscillator section. This circuit comes into action when mains voltage is not there. It, along with two banks of transistors T1-T8, will generate low-level AC voltage (15V-0-15V) at terminals of transformer X2, which will be stepped up by transformer X2.

The circuit is built around NE555 timer (IC2), dual JK flip-flop 4027 (IC1), transistors SK100 (T11-T12) and BC547 (T9-T10), voltage regulator 7805 (IC3) and a few other components. NE555 timer is configured in astable multivibrator mode.

Frequency of the timer is set to around 200Hz with the help of preset VR1 in order to get around 50Hz line frequency at CON1. Output of the timer from its pin 3 (CP2) of second flip-flop of IC1 as clock pulse. Output of this flip-flop from pin 1 (Q2) is used to clock the first flip-flop. Outputs Q1 and Q1

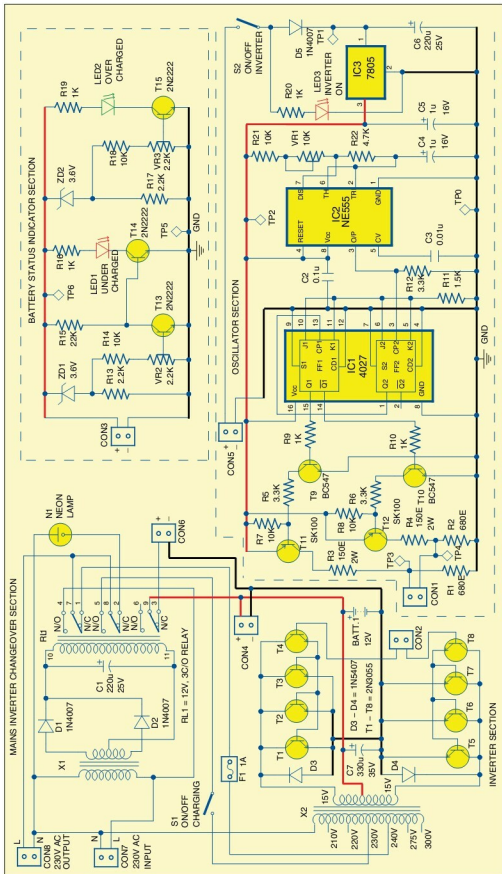


Fig. 1: Circuit diagram of the mini UPS